

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
imgorbunov@imgorbunov:~$ ls /etc/ > file.txt
imgorbunov@imgorbunov:~$ ls >> file.txt
imgorbunov@imgorbunov:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
```

Рисунок 1: Запись в файл

```
imgorbunov@imgorbunov:~$ grep .conf file.txt > conf.txt
imgorbunov@imgorbunov:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
```

Рисунок 2: Поиск расширения .conf


```
0072000700000000
/home/imgorbunov/.npm/_cacache/index-v5/cd
/home/imgorbunov/.npm/_cacache/index-v5/c9
/home/imgorbunov/.npm/_cacache/index-v5/c0
/home/imgorbunov/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea3
10cda0b203217f693
/home/imgorbunov/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d13
50482a1d990504757
/home/imgorbunov/.npm/_cacache/index-v5/af/cf
/home/imgorbunov/.npm/_cacache/index-v5/af/3b/c01fad9ea9ac6256e90d5a59e2c8b816cc7bcf0f41a
5ca46eaa0bbab8743
/home/imgorbunov/.npm/_cacache/index-v5/c3
/home/imgorbunov/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f2506
214f94db8861160d3
/home/imgorbunov/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c81
88a44d554f3f3ba2e
/home/imgorbunov/.npm/_cacache/index-v5/ca
/home/imgorbunov/.npm/_cacache/index-v5/95/f7/cabf02718d2ced903e42baa9d89ccac492bdd331ae8
0e550538ac79b0d3a
/home/imgorbunov/snap/hugo/common
/home/imgorbunov/snap/hugo/current
/home/imgorbunov/conf.txt
imgorbunov@imgorbunov:~$
```

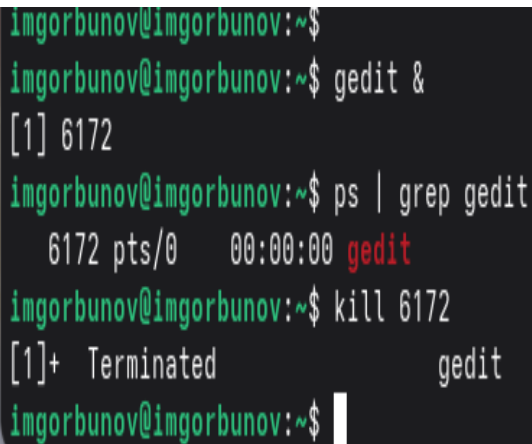
Рисунок 3: Поиск файлов

```
/etc/foomatic/hashes.d/hashes.new
find: '/etc/libvirt'/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
: Permission denied
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
:find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
find: '/etc/ssh/ssh_config.d': Permission denied
find: '/etc/sssd': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
imgorbunov@imgorbunov:~$  
imgorbunov@imgorbunov:~$ find ~ -name "log*" > logfile &  
[1] 6134  
imgorbunov@imgorbunov:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
imgorbunov@imgorbunov:~$ rm logfile  
imgorbunov@imgorbunov:~$
```

Рисунок 5: Фоновый запуск процесса



```
imgorbunov@imgorbunov:~$  
imgorbunov@imgorbunov:~$ gedit &  
[1] 6172  
imgorbunov@imgorbunov:~$ ps | grep gedit  
  6172 pts/0    00:00:00 gedit  
imgorbunov@imgorbunov:~$ kill 6172  
[1]+  Terminated                  gedit  
imgorbunov@imgorbunov:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                     User Commands                               DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of
    space available on the file system containing each file name argument.  If no
    file name is given, the space available on all currently mounted file systems is
    shown.  Space is shown in 1K blocks by default, unless the environment variable
    POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted
    file system, df shows the space available on that file system rather than on the
    file system containing the device node.  This version of df cannot show the space
    available on unmounted file systems, because on most kinds of systems doing so
    requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file
    systems by default.

    Mandatory arguments to long options are mandatory for short options too.

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory, for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size
        is usually smaller, it may be larger due to holes in ('sparse') files, in-
        ternal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
imgorbunov@imgorbunov:~$ df
imgorbunov@imgorbunov:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 89581524 64935404  58% /
devtmpfs         4011284      0    4011284   0% /dev
tmpfs            4035660      92    4035568   1% /dev/shm
tmpfs            1614264     1952    1612312   1% /run
tmpfs            1024         0      1024      0% /run/credentials/systemd-journald.servic
e
tmpfs            4035660     1644    4034016   1% /tmp
/dev/nvme0n1p3 155186176 89581524 64935404  58% /home
/dev/nvme0n1p2  1992552    518800    1352512  28% /boot
/dev/loop1        68480      68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2        49280      49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0       115712     115712         0 100% /var/lib/snapd/snap/hugo/25676
tmpfs             1024         0      1024      0% /run/credentials/systemd-resolved.servic
e
tmpfs             807132      164    806968   1% /run/user/1027
/dev/sr0         2677920    2677920         0 100% /run/media/imgorbunov/Fedora-WS-Live-43
imgorbunov@imgorbunov:~$
```

Рисунок 9: Справка по команде du

```
424      ./npm/_cacache/index-v5
115472   ./npm/_cacache
115536   ./npm
16       ./snap/hugo/25676/.config/go/telemetry/local
0        ./snap/hugo/25676/.config/go/telemetry/upload
16       ./snap/hugo/25676/.config/go/telemetry
16       ./snap/hugo/25676/.config/go
16       ./snap/hugo/25676/.config
16       ./snap/hugo/25676
0        ./snap/hugo/common
20       ./snap/hugo
20       ./snap
0        ./monthly
0        ./reports/monthly/monthly
0        ./reports/monthly
0        ./reports
4        ./ski.plases/equipment
0        ./ski.plases/plans
4        ./ski.plases
0        ./australia
0        ./play/games/play
0        ./play/games
0        ./play
1552236  .
imgorbunov@imgorbunov:~$
```

Рисунок 10: Запуск команды du


```
/home/imgorbunov/.npm/_cacache/index-v5/35/17  
/home/imgorbunov/.npm/_cacache/index-v5/30  
/home/imgorbunov/.npm/_cacache/index-v5/30/1a  
/home/imgorbunov/snap  
/home/imgorbunov/snap/hugo  
/home/imgorbunov/snap/hugo/25676  
/home/imgorbunov/snap/hugo/25676/.config  
/home/imgorbunov/snap/hugo/25676/.config/go  
/home/imgorbunov/snap/hugo/25676/.config/go/telemetry  
/home/imgorbunov/snap/hugo/25676/.config/go/telemetry/local  
/home/imgorbunov/snap/hugo/25676/.config/go/telemetry/upload  
/home/imgorbunov/snap/hugo/common  
/home/imgorbunov/monthly  
/home/imgorbunov/reports  
/home/imgorbunov/reports/monthly  
/home/imgorbunov/reports/monthly/monthly  
/home/imgorbunov/ski.places  
/home/imgorbunov/ski.places/equipment  
/home/imgorbunov/ski.places/plans  
/home/imgorbunov/australia  
/home/imgorbunov/play  
/home/imgorbunov/play/games  
/home/imgorbunov/play/games/play  
imgorbunov@imgorbunov:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.